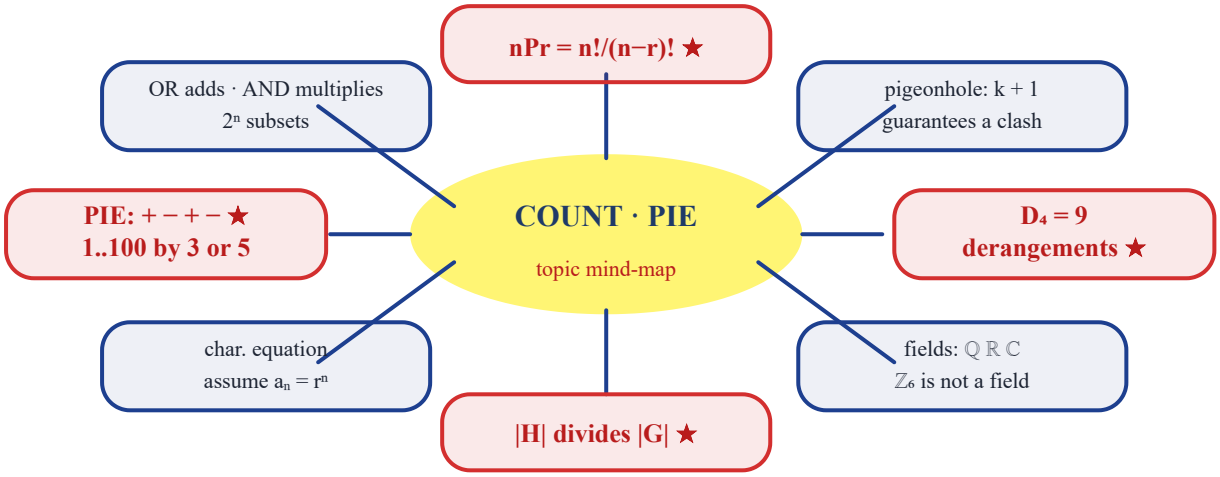
Counting & Algebraic Structures

Inclusion–exclusion · derangements · recurrences · groups & rings: 11 sheets



↑ the whole topic on one page: red = high-yield, blue = definitions

Syllabus coverage

"… Mathematical reasoning: induction, contradiction and recursion; **Counting; Principles of inclusion and exclusion; Recurrence relations; Algebraic structures: rings and groups**." Those four items are this notebook; sets, relations, induction and graph theory belong to other lessons of the chapter.

High-yield list

① Addition = **OR**, multiplication = **AND**

② n! arrangements, **0! = 1**, subsets = **2^n**

③ ${}^nP_r = n!/(n-r)!$, ${}^nC_r = n!/[r!(n-r)!]$

④ **nCr = nPr / r!**

⑤ LEVEL ⇒ $5!/(2!\cdot 2!) = 30$

⑥ $|A \cup B| = |A|+|B|-|A\cap B|$; 3-set adds the triple back

⑦ 1..100 by 3 or 5 = 33+20−6 = **47**

⑧ Pigeonhole on months ⇒ **13**

⑨ $D_n = n!(1-1+1/2!-1/3!+\ldots)$; **$D_4 = 9$**

⑩ Fib. ⇒ $r^2 - r - 1 = 0$; $a(n) = a(n-1)+6a(n-2)$ ⇒ r = 3, −2

⑪ Lagrange: **|H| divides |G|**

⑫ Field = commutative ring, every non-zero invertible

Legend: **yellow** = key term · **circle** = memorise · purple = trick · red = trap · green = example

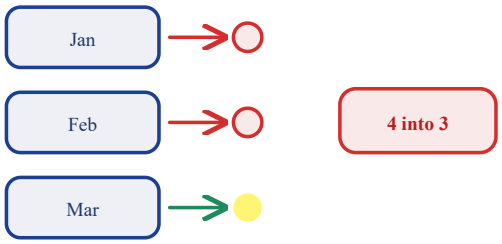
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1 Counting Principles, Subsets & Pigeonhole

Definition: **Counting** (combinatorics) is the study of **how many** ways a collection of objects can be selected, arranged or distributed. **n! = n(n−1)⋯1** counts the ways of putting n distinct objects in a row, and **0! = 1** — the one empty arrangement.

Principle	Exact statement	Use it when
Addition (Sum)	a task is done in m ways **or** n ways, never both ⇒ **m + n**	the cases are **mutually exclusive**
Multiplication (Product)	a task needs m ways **and** then n ways ⇒ **m × n**	the stages happen **one after another**
Bijection	two sets have the same size iff a **one-to-one pairing** exists	you can **rename** one set as the other



more birds than holes ⇒ a clash

pigeonhole: objects > boxes forces a collision

The pigeonhole principle (Dirichlet): Put n objects into k boxes. **If n > k, at least one box holds two or more.** A "guarantee" question therefore answers **n = k + 1**, and some box gets at least **⌈n/k⌉**.

Worked example ✍ birthdays: Months are the boxes, k = 12. With 12 students every one **could** be born in a different month; with **13** a repeat is forced ⇒ **13**. The day version needs k = 366 and gives 367.

Memory trick: "**O**R **A**dds, **A**ND **M**ultiplies." Read the word joining the two conditions: **or** means add the two counts, "must happen together" means multiply.

Exam trap: (i) Addition needs **disjoint** cases; overlapping ones need inclusion–exclusion (sheet 4). (ii) **n! counts order** — 5 people in 5 chairs is 5! = **120**; 720 is 6!, the trap. (iii) 0! = **1**. (iv) For a pigeonhole guarantee, "12 months" is **not** the answer.

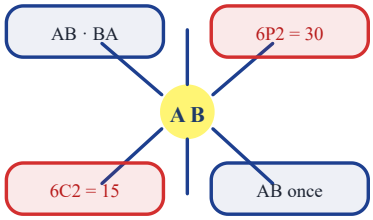
Worked example ✍ subsets and power sets: A set with n elements has exactly **2^n subsets**, because every element is independently in or out. It has **$2^n - 1$** proper subsets and **$2^n - 1$** non-empty subsets — the two counts coincide. A 5-element set therefore has 32, 31 and 31. **Splitting a set into two labelled parts is also 2^n.**

IT link: Binary representation of a k-bit string has **2^k** values, and the addition principle is literally how address ranges are summed: a /24 block has 2^8 = **256** addresses, and after the network and broadcast addresses are reserved **254** remain for hosts. Choosing a bitmask with n set bits out of 32 is ${}^{32}C_n$, which is why subnet masks are counted that way. Pigeonhole reappears in hashing: n keys in m buckets forces a collision as soon as **n > m**, which is the entire reason hash tables have a load factor.

2 Permutations & Combinations

Definition: An **arrangement** of r objects chosen from n distinct objects, **in which order matters**, is a **permutation**: $^{n}P_{r} = n! / (n - r)!$. When order **does not** matter it is a **combination**: $^{n}C_{r} = n! / [r!\,(n - r)!]$. Each combination generates exactly r! orderings of its members, so $^{n}C_{r} = {}^{n}P_{r} / r!$, and $^{n}C_{r} = {}^{n}C_{n-r}$.

Question type	Tool	Watch out
seating, ranking, lining up, codes without repetition	$^{n}P_{r}$	order matters
committee, team, hand of cards, choose r dates	$^{n}C_{r}$	order irrelevant
select all r, then arrange them	$^{n}C_{r} \cdot r!$	gives back $^{n}P_{r}$
objects repeated p, q, … times	$n! / (p!\,q!\,\ldots)$	divide by **every** repeat
circular seating of n people	$(n - 1)!$	rotations are identical



30 ÷ 2! = 15 — committee

two chosen from six: order on, order off

Memory trick: "**P**ermutation **P**ositions — order is a position, so it counts; **C**ombination is a **C**ommunity — no chair, no order." Divide the permutation answer by **r!** to strip the order off: that is the size of the error.

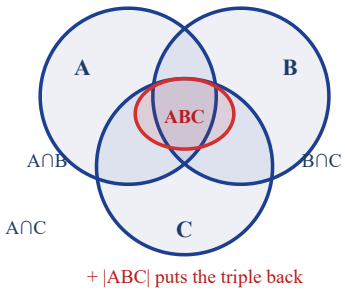
Worked examples ✍️: **(a)** Committee of 2 from 6: order irrelevant ⇒ $^{6}C_{2} = 15$. **30** = $^{6}P_{2}$ counts **ordered** pairs, the standard distractor. **(b)** 5 books, 2 chosen and placed: $^{5}P_{2} = 20$; "which two" alone is $^{5}C_{2} = 10$. **(c)** 5 people in 5 chairs = 5! = **120**; round a table (5−1)! = 24.

Worked example ✍️ letters with repeats: **LEVEL**: 5 letters, L twice and E twice, so distinct arrangements = **5!/(2!·2!) = 120/4 = 30**. Divide once per repeated letter in **BOOKKEEPER**, and check the quotient is a **whole number**.

Exam trap: (i) $^{n}C_{r} = {}^{n}P_{r}/r!$, not /n!. (ii) At r = n the combination is 1 and the permutation is n!. (iii) "Round a table" removes rotations, so divide by n. (iv) Two identical coins **are** indistinguishable; two people who merely look alike are not. (v) A password is a permutation even when digits repeat.

3 Principle of Inclusion–Exclusion (PIE)

Definition: The **principle of inclusion–exclusion** counts the elements of a union of overlapping sets. For two sets $|A \cup B| = |A| + |B| - |A \cap B|$: adding the two sizes counts the overlap twice, so subtract it once. For three sets $|A \cup B \cup C| = |A|+|B|+|C| - |A\cap B| - |A\cap C| - |B\cap C| + |A\cap B\cap C|$ — the triple was added once and subtracted three times, so it must be **added back once**.



subtract each pairwise overlap, add the triple back

Quantity you want	Formula
none of the sets	$\lvert U\rvert - \lvert A \cup B \cup C\rvert$
exactly one set	$\Sigma\lvert\text{single}\rvert - 2\,\Sigma\lvert\text{pair}\rvert + 3\,\lvert\text{triple}\rvert$
at least one	$\lvert U\rvert - \lvert A' \cap B' \cap C'\rvert$
at least two	$\Sigma\lvert\text{pair}\rvert - 2\,\lvert\text{triple}\rvert$

General alternating form: $|A_1 \cup \ldots \cup A_n| = \Sigma|A_i| - \Sigma|A_i \cap A_j| + \Sigma|A_i \cap A_j \cap A_k| - \cdots + (-1)^{n-1}\Sigma_{all}\,A_i$. The signs strictly alternate +, −, +, −; the k-th term combines **k** sets at a time.

Worked example ✍ integers 1 to 100 divisible by 3 or by 5: Let A = multiples of 3, B = multiples of 5. $|A| = \lfloor 100/3 \rfloor$ = **33**, $|B| = \lfloor 100/5 \rfloor$ = **20**, A ∩ B = multiples of 15 so $|A \cap B| = \lfloor 100/15 \rfloor$ = **6**. Hence $|\mathbf{A \cup B| = 33 + 20 - 6 = 47}$. Trap options: 33 alone, 53 = 33 + 20 with no subtraction, 41 = 33 + 20 − 12.

Worked example ✍ none of three subjects: 100 students; 45 read M, 40 read P, 35 read C; 15 read M and P, 10 read P and C, 12 read M and C; 5 read all three. $|M \cup P \cup C| = 45 + 40 + 35 - 15 - 10 - 12 + 5 = 120 - 37 + 5 =$ **88**, so those reading **none** = 100 − 88 = **12**. Forgetting to add the 5 back gives 17, the standard wrong option.

Memory trick: "**P**lus the singles, **m**inus the pairs, **p**lus the triples", then keep flipping the sign. An element in exactly **k** sets must finish counted once, and the alternating sum $1 - (k-1) + (k-2 \text{ choose } 2) - \cdots = (1-1)^k = 0$ guarantees it.

Exam trap: (i) "At least one" is a **union**; "none" is its complement. (ii) The triple intersection is added **once** — dropping it is the commonest error. (iii) Write the six pairwise terms out before substituting. (iv) A universal set of size n is **assumed** unless stated.

Worked example ✍ exactly one subject: In the class above, exactly **one** subject = 45 + 40 + 35 − 2(15 + 10 + 12) + 3(5) = 120 − 74 + 15 = **61**. Exactly **two** = 15 + 10 + 12 − 3(5) = **22**, and 61 + 22 + 5 = 88 with the 12 outside making 100 — a useful internal check.

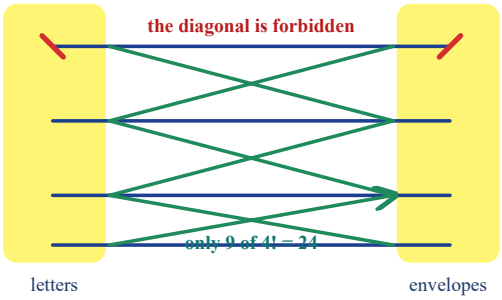
Counting by complements — the habit that saves time:

When a question says **at least one**, compute **none** and subtract, because the "none" count is always the easier one. "Not both" = total − both; "at least one of each kind" = total − (no A) − (no B) + (no A and no B).

4 PIE in Depth & Derangements

Definition: A **derangement** of n objects is a permutation in which **no object is in its own place**. Let D_n count them. Inclusion–exclusion applied to the n events "object i is in its place" gives $\mathbf{D_n = n!\,(1 - 1/1! + 1/2! - 1/3! + \cdots + (-1)^n/n!)}$, and an equivalent two-step recurrence is $\mathbf{D_n = (n-1)(D_{n-1} + D_{n-2})}$.

n	n!	D_n by the formula	D_n by $(n-1)(D_{n-1}+D_{n-2})$
1	1	0	—
2	2	1	$1 \times (1 + 0) = 1$
3	6	2	$2 \times (1 + 1) = 2$
4	24	**9**	$3 \times (2 + 1) = 9$
5	120	44	$4 \times (9 + 2) = 44$
6	720	265	$5 \times (44 + 9) = 265$



the red cross kills the four fixed points

Worked example ✍️ four letters into four addressed envelopes: $D_4 = 4!\,(1 - 1 + 1/2 - 1/6 + 1/24) = 0 + 24/2 - 24/6 + 24/24 = 0 + 12 - 4 + 1 = \mathbf{9}$. Quick route: $D_4 = 3(D_3 + D_2) = 3 \times (2 + 1) = \mathbf{9}$. Near misses: 6, 11, 23.

How to read the derangement formula: Start from all **n!** arrangements and apply PIE to the "in its place" events: subtract the arrangements with a specified letter home, add back the pairs that are both home, subtract the triples, and so on to the n-th sign. The events are symmetric, so each k-fold intersection contains **(n−k)!** arrangements — that is exactly where the 1/k! terms come from.

Exam trap: (i) The first two terms **cancel**: $n! - n!/1! = 0$. (ii) $D_1 = \mathbf{0}$, which is why the recurrence needs two seeds. (iii) For $n \geq 2$, D_n is never more than **n!/2**. (iv) "No two letters adjacent" is a different PIE problem with blocks, and the plain formula will not give it.

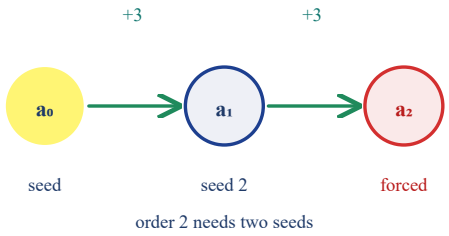
IT link: A spell checker or a firewall rule base avoids double counting: a packet rejected by several rules must be subtracted back into the "rejected at least once" count. The same arithmetic decides how many inputs a **k-of-n** threshold circuit accepts.

Worked example ✍️ a derangement variant: "Nobody in place, and A must not be first." Split on where A goes: it can go to position 2 or 3, and each branch contributes $D_3 + D_2 = 3$, so the answer is $2 \times 3 = \mathbf{6}$. This case split on one object is the standard attack on any restricted-arrangement question. **Stirling numbers** $S(n, k)$ — partitions of n objects into k non-empty unordered groups — refine this further: $S(n, 2) = 2^{n-1} - 1$.

5 Recurrence Relations: Definition & Types

Definition: A **recurrence relation** is an equation expressing $\mathbf{a_n}$ in terms of a fixed number of its **preceding terms**. The number of preceding terms used is the **order**: $a_n = a_{n-1} + 6a_{n-2}$ is of **order 2**. The independent values are the **initial conditions**, and **a recurrence of order k needs exactly k initial values** to fix the whole sequence.

Classification	Test	Example
Linear	a_n appears to the **first power** only	$a_n = 2a_{n-1} + 3$
Non-linear	a_n multiplied by itself or to a variable power	$a_n = a_{n-1}^2$, $a_n = n \cdot a_{n-1}$
Homogeneous	no term depending on **n alone**	$a_n = a_{n-1} + a_{n-2}$
Non-homogeneous	a term such as 3, n or 2^n appears	$a_n = 2a_{n-1} + n$
Order 1 / order 2	one / two preceding terms	$a_n = 2a_{n-1} + 1$ / Fibonacci
Constant coefficients	the multipliers of a_{n-1}, a_{n-2} … do not vary with n	$a_n = 3a_{n-1} - 2a_{n-2}$



iteration: each arrow is one application, never a division

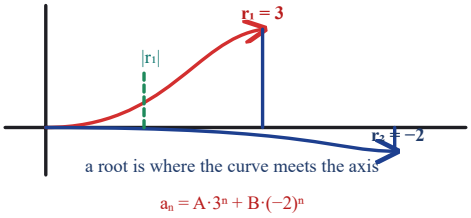
The three solving methods: **1. Iteration** — write a_1, a_2, a_3 … and read the pattern. **2. Characteristic equation** — for a linear homogeneous relation with constant coefficients; the standard method and the one that carries marks. **3. Generating function** — replace a_n by $A(x) = \Sigma\, a_n x^n$, multiply by x^n and sum, then read off a rational function whose coefficient expansion is the sequence.

Worked example ✍ iteration: $a_n = a_{n-1} + 3$ with $a_0 = 2$ gives 5, 8, 11 — each step adds the **same** 3, so $a_n = \mathbf{2 + 3n}$. For $a_n = 2a_{n-1} + 3$ with $a_0 = 1$ the values 5, 13, 29 **double then add 3**, which is not geometric, so shift: set $b_n = a_n + 3$, then $b_n = 2b_{n-1}$ and $\mathbf{a_n = 4 \cdot 2^n - 3}$.

Exam trap: (i) "Homogeneous" is about the **n-dependent term** — $a_n = n \cdot a_{n-1}$ is non-homogeneous. (ii) Order counts the **highest lag**, so $a_n = a_{n-2}$ is still order 2. (iii) Iteration is only a legal written method if the closed form is then **proved** by induction. (iv) A repeated root needs the extra n factor (see sheet 7).

6 Characteristic-Equation Method

Method: For $\mathbf{a_n = c_1 a_{n-1} + c_2 a_{n-2} + \cdots + c_k a_{n-k}}$ substitute the trial solution $\mathbf{a_n = r^n}$. Dividing by r^{n-k} gives the **characteristic equation** $\mathbf{r^k - c_1 r^{k-1} - c_2 r^{k-2} - \cdots - c_k = 0}$. Each **distinct** root r_i contributes $A_i r_i^n$; a root repeated m times contributes $r^n(A + Bn + \cdots)$. The constants come from the initial conditions.



each root gives one term of the general solution

Worked example ✍ **$a(n) = a(n-1) + 6a(n-2)$, $a_0 = 2$, $a_1 = 1$: Step 1 — type.** Order 2, linear, homogeneous, constant coefficients ⇒ the characteristic method applies. **Step 2 — characteristic equation.** Put $a_n = r^n$: $r^n = r^{n-1} + 6r^{n-2}$ ⇒ $\mathbf{r^2 = r + 6}$ ⇒ $\mathbf{r^2 - r - 6 = 0}$ ⇒ $\mathbf{(r-3)(r+2) = 0}$, so $r_1 = 3$, $r_2 = -2$, distinct. **Step 3 — general solution.** $\mathbf{a_n = A\cdot 3^n + B\cdot(-2)^n}$. **Step 4 — initial conditions.** n = 0: A + B = 2. n = 1: 3A − 2B = 1. Twice the first plus the second gives 5A = 5 ⇒ A = 1, and B = 1. **Step 5 — check.** $\mathbf{a_n = 3^n + (-2)^n}$. From the recurrence $a_2 = 1 + 6\cdot 2 =$ **13**, and the formula gives 9 + 4 = 13 ✓. $a_3 = 13 + 6\cdot 1 =$ **19**, and the formula gives 27 − 8 = 19 ✓.

Recurrence	Characteristic equation	Roots
Fibonacci $F_n = F_{n-1} + F_{n-2}$	$r^2 - r - 1 = 0$	$(1 \pm \sqrt{5})/2$
$a_n = a_{n-1} + 6a_{n-2}$	$r^2 - r - 6 = 0$	3, −2
$a_n = 2a_{n-1} + a_{n-2}$	$r^2 - 2r - 1 = 0$	$1 \pm \sqrt{2}$
$a_n = 3a_{n-1}$ (order 1)	$r - 3 = 0$	3
$a_n = 2a_{n-1} - a_{n-2}$ (double root)	$(r-1)^2 = 0$	1 twice ⇒ A + Bn

Two sequences worth memorising: **Arithmetic** — $a_n = a_{n-1} + d$, giving $\mathbf{a_n = a + nd}$, with $\Sigma_{k=1}^{n} a_k = n/2\cdot(2a + (n-1)d)$. **Geometric** — $a_n = r\cdot a_{n-1}$, giving $\mathbf{a_n = a\cdot r^n}$, with $\Sigma_{k=1}^{n} a_k = a(r^n - 1)/(r - 1)$ for $r \neq 1$.

Exam trap: (i) Fibonacci's characteristic equation is $\mathbf{r^2 - r - 1 = 0}$; $r^2 - r + 1 = 0$ is the classic slip. (ii) Move everything to one side first: $a_n = a_{n-1} + 6a_{n-2}$ gives $r^2 - r - 6$, **not** $r^2 - r + 6$. (iii) A repeated root needs $(A + Bn)r^n$, never Ar^n. (iv) Order 2 means **two** seeds.

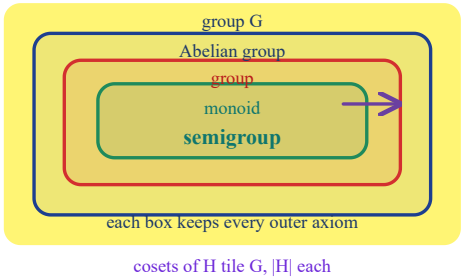
Where recurrences earn their keep: Master-s Theorem analyses divide-and-conquer algorithms. For $T(n) = 2T(n/2) + n$: a = 2, b = 2, so $n^{\log_b a} = n$, and f(n) = n is Θ(n) — **case 2**, giving $\mathbf{T(n) = \Theta(n \log n)}$. Case 1 is when f(n) dominates (merge sort), case 3 when it is polynomially smaller (binary search). Fibonacci is the cost of naive recursion, so an app reporting "how many calls" is solving a recurrence. Dynamic-programming tables (edit distance, coin change) are the same idea written bottom-up.

7 Semigroups, Groups & Lagrange's Theorem

Definition: A **binary operation** * on a set G assigns to every ordered pair (a, b) exactly one element of G. The four **group axioms** are **closure** (a*b ∈ G), **associativity** ((a*b)*c = a*(b*c)), **identity** (∃e with a*e = e*a = a) and **inverse** (∀a, ∃a^{-1} with $a*a^{-1} = e$). Add **a*b = b*a** and the group is **Abelian**.

Associativity and commutativity are different properties.

Structure	What it adds
Semigroup	closure + associativity only
Monoid	a semigroup with an identity element
Group	a monoid in which every element has an inverse
Abelian group	a group that is also commutative
Cyclic group	generated by one element g: $\{g^0, g^1, \ldots\}$
Subgroup $H \leq G$	a subset that is itself a group under the same operation



one axiom per step; cosets cut the group evenly

Lagrange's theorem: Let H be a subgroup of a finite group G. Then **|H| divides |G|**, and |G| = |H| × (number of distinct cosets). In words: **the order of a subgroup must divide the order of the group**, and the cosets cut G into equal pieces. Corollary: a group of prime order p has **no non-trivial proper subgroup**.

Worked example ✍️: **$(\mathbb{Z}, +)$** is an **Abelian group**: closure, associativity, identity **0**, inverse **−a**, commutative. **$(\mathbb{Z}_6, +)$** is a **cyclic** Abelian group of order 6 generated by 1; by Lagrange it has subgroups of order 1, 2 and 3 only. **$(\mathbb{Q}^*, \times)$** is an Abelian group with identity 1, but **$(\mathbb{Z}^*, \times)$** is not, since 2 has no inverse in $\mathbb{Z}$.

Exam trap: (i) Lagrange is **necessity, not sufficiency**: 4 divides 12, but a group of order 12 need not contain a subgroup of order 4 (A_4 does not). (ii) In $(\mathbb{Z}_6, +)$ the element 2 has order 3 and 3 has order 2. (iii) A **semigroup has no identity**. (iv) Closure is not automatic: on $\mathbb{Z}$ the operation − is **not** closed, since $1 - 2 = -1 \notin \mathbb{Z}$.

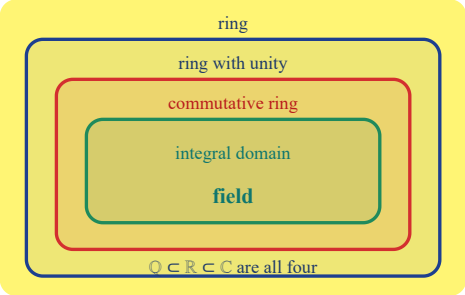
Order of an element, two groups to recognise, and the crypto link: The **order of g** is the least positive k with $g^k = e$; in $\mathbb{Z}_n$ it is n/gcd(n, g), so ord(2) in $\mathbb{Z}_6$ is 3 and ord(3) is 2. The **symmetric group S_n** of all permutations of n objects is a group of order n! — not Abelian for $n \geq 3$, because composition reverses order. The **alternating group A_n** is the subgroup of even permutations and has order n!/2, exactly as Lagrange demands. Every public-key cipher is a group in disguise: encryption is an element of a finite group, decryption is its inverse, and security means nobody can compute that inverse without the private key. RSA works in $\mathbb{Z}_n^*$, of order φ(n).

8 Rings, Integral Domains & Fields

Definition: A **ring** (R, +, ×) has two binary operations: (R, +) is an **Abelian group**, (R, ×) is a **semigroup** (closed and associative, identity not required), and multiplication **distributes** over addition both ways. Add a multiplicative identity 1 and it is a **ring with unity**; add commutativity of × and it is **commutative**; add "no zero divisors" and it is an **integral domain**; add "every non-zero element is invertible" and it is a **field**.

Set with operations	Abelian group?	Ring with unity?	Integral domain?	Field?
$(\mathbb{Z}, +, \times)$	yes, under +	**yes**	**yes**	**no**
$(\mathbb{Q}, +, \times)$	yes, under +	**yes**	**yes**	**yes**
$(\mathbb{R}, +, \times)$	yes, under +	**yes**	**yes**	**yes**
$(\mathbb{C}, +, \times)$	yes, under +	**yes**	**yes**	**yes**
$(\mathbb{Z}_6, +, \times)$	yes, under +	**yes**	**no**	**no**
$M_2(\mathbb{R})$, matrix + and ×	yes, under +	**yes**	**no**	**no**
GL(n), × only	yes; Abelian only $n \leq 2$	n/a	n/a	n/a

Why each row reads that way: $\mathbb{Z}$ fails only the field test: 2 has no inverse in $\mathbb{Z}$. $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ are fields — $1/z$ is the conjugate of z scaled by $|z|^2$. $\mathbb{Z}_6$ has zero divisors ($2 \cdot 3 = 0$) and $\mathbf{M_2(\mathbb{R})}$ contains a zero matrix, so neither is a domain; **GL(n)** is settled by the **determinant**.



each inner box adds exactly one condition

Memory trick: "**G**roup = **one** operation, and it must have an **I**nverse; **R**ing = **two** operations, and multiplication need **not** have one." Read the hierarchy downward as **F**ield → **I**ntegral **D**omain → **C**ommutative ring → ring → semigroup: the extra letters are exactly the extra conditions.

Exam trap: (i) A **ring need not have an identity** — that is the whole difference between "ring" and "ring with unity". (ii) $\mathbb{Z}$ is an integral domain but **not** a field: only ±1 are units. (iii) 6 being composite is not why $\mathbb{Z}_6$ fails; the zero divisors are. (iv) Every ring is an **Abelian group under +**.

Applications: **Cryptography.** RSA works in $\mathbb{Z}_n$ under multiplication: the public key uses e with $\gcd(e, \varphi(n)) = 1$, the private key d with $ed \equiv 1 \bmod \varphi(n)$. **Elliptic-curve cryptography** swaps that multiplicative group for the points on a curve — still a group, with far smaller keys. **Coding theory** builds linear codes as subgroups of $\mathbb{Z}_2^n$, measured by minimum Hamming distance: correcting t errors needs $d \geq 2t + 1$. **Computer algebra** works over fields so a non-zero polynomial has at most its degree many roots, which is what makes gcd and integration terminate.

Night-Before-the-Exam Sheet

Fact sheet:

OR adds, AND multiplies	n! arrangements; **0! = 1**
n-element set ⇒ $\mathbf{2^n}$ subsets	Proper subsets = $2^n - 1$
Pigeonhole: $n > k$ ⇒ a clash	Guarantee needs **k + 1**
Months ⇒ **13**; days ⇒ 367	${}^nP_r = n!/(n-r)!$
${}^nC_r = n!/[r!(n-r)!]$	**nCr = nPr/r!**
$nCr = nC_{n-r}$	Repeats: $n!/(p!\,q!\,\ldots)$
LEVEL ⇒ $5!/(2!2!) = \mathbf{30}$	Round a table ⇒ $(n-1)!$
$\lvert A \cup B\rvert = \lvert A\rvert+\lvert B\rvert-\lvert A\cap B\rvert$	3-set: add the triple back
1..100 by 3 or 5 = **47**	None of three = **12**
$D_n = n!(1-1+1/2!-1/3!+\ldots)$	$\mathbf{D_4 = 9}$, $D_5 = 44$, $D_6 = 265$
$D_n = (n-1)(D_{n-1}+D_{n-2})$	Order k ⇒ k seeds
Arithmetic $a_n = a + nd$	Geometric $a_n = a \cdot r^n$
Fibonacci ⇒ $r^2 - r - 1 = 0$	$a(n)=a(n-1)+6a(n-2)$ ⇒ $r = 3, -2$
Solution $a_n = \mathbf{3^n + (-2)^n}$	$a_2 = 13$, $a_3 = 19$ ✓
Repeated root ⇒ $(A + Bn)r^n$	Lagrange: $\lvert H\rvert$ **divides** $\lvert G\rvert$
Semigroup ⊂ monoid ⊂ group	Field = commutative ring + inverses
$\mathbb{Z}$ is a domain, **not** a field	$\mathbb{Z}_6$ has zero divisors
$M_2(\mathbb{R})$ is a ring, not a domain	GL(n) under × is a group

Top 12 one-liners for MCQs

1. Order matters ⇒ **nPr**, not nCr
2. **nCr = nPr / r!**
3. Three-set PIE **adds** the triple back
4. "At least one" is a **union**
5. Derangements: $\mathbf{D_4 = 9}$
6. Pigeonhole guarantee = **holes + 1**
7. Homogeneous ⇒ no **n-only** term
8. Fibonacci ⇒ $\mathbf{r^2 - r - 1 = 0}$
9. Order 2 ⇒ **two** initial conditions
10. Lagrange is **necessity only**
11. A ring need **not** have 1
12. $\mathbb{Z}$ is a domain, **not** a field

Top traps: "5 people in 5 chairs = 720" (that is 6!) · reading a committee question as ordered, so 30 instead of 15 · forgetting the + $\lvert A\cap B\cap C\rvert$ term and getting 17 instead of 12 · answering "12 months" to a pigeonhole guarantee question · forgetting the 2!·2! divisor and getting 120 for LEVEL · treating D_n as $n! - n$ · writing $r^2 - r + 6$ for $a(n) = a(n-1) + 6a(n-2)$ · using one seed for an order-2 recurrence · quoting Lagrange as "every divisor appears" · calling $\mathbb{Z}$ a field.

COUNT RIGHT
nCr = nPr/r! · PIE signs alternate
char. equation · Lagrange

COUNT WRONG
720 for 5! · 30 for a committee
12 for pigeonhole · $\mathbb{Z}$ a field

every wrong option in the bank is one of these

the whole lesson reduces to eight numbers and four traps

✓ Practice MCQs (exam pattern, 10 marks)

1. 5 people in a row of 5 chairs:
(a) 120 (b) 720
(c) 60 (d) 24

2. A committee of 2 from 6 people:
(a) 30 (b) 15
(c) 12 (d) 36

3. Integers 1..100 divisible by 3 or 5:
(a) 53 (b) 41
(c) 47 (d) 33

4. Students to guarantee two born in one month:
(a) 12 (b) 24
(c) 366 (d) 13

5. Distinct arrangements of LEVEL:
(a) 30 (b) 60
(c) 120 (d) 20

6. Four letters into four envelopes, none correct:
(a) 11 (b) 23
(c) 6 (d) 9

7. 100 students, 45 M / 40 P / 35 C, 15 MP / 10 PC / 12 MC, 5 all — study none:
(a) 17 (b) 12
(c) 22 (d) 7

8. Subsets of a 5-element set:
(a) 31 (b) 16
(c) 32 (d) 25

9. Characteristic equation of Fibonacci:
(a) $r^2 - r - 1 = 0$ (b) $r^2 + r - 1 = 0$
(c) $r^2 - r + 1 = 0$ (d) $r^2 - 2r - 1 = 0$

10. A group of order 12 must have a subgroup of order:
(a) 5 (b) 7
(c) 4 (d) 8

Answer key → 1-a · 2-b · 3-c · 4-d · 5-a · 6-d · 7-b · 8-c · 9-a · 10-c

Written question: (i) Solve $a(n) = a(n-1) + 6a(n-2)$, $n \geq 2$, $a(0) = 2$, $a(1) = 1$, verifying at $n = 2, 3$. (ii) How many integers 1..100 are divisible by 3 or 5, and by neither? (iii) Classify $(\mathbb{Z}, +, \times)$, $(\mathbb{Z}_6, +, \times)$, $M_2(\mathbb{R})$, $(\mathbb{Q}, +, \times)$. (iv) State Lagrange's theorem and list the possible subgroup orders of a group of order 12. 10

1. **Recurrence (3)** — $a_n = r^n$ gives $r^2 - r - 6 = 0 \Rightarrow r = 3, -2$. Then $A + B = 2$ and $3A - 2B = 1 \Rightarrow A = B = 1$, so $\mathbf{a_n = 3^n + (-2)^n}$. Check $a_2 = 13 = 9 + 4$ ✓, $a_3 = 19 = 27 - 8$ ✓. 3
2. **Inclusion–exclusion (2)** — $33 + 20 - 6 \Rightarrow$ **47** by 3 or 5, **53** by neither. 2
3. **Classification (3)** — $\mathbb{Z}$: domain, **not** a field. $\mathbb{Z}_6$ and $M_2(\mathbb{R})$: rings with unity with zero divisors, so neither domain nor field. $\mathbb{Q}$: all four. Every ring is an **Abelian group under +**. 3
4. **Lagrange (2)** — **|H| divides |G|**; for |G| = 12 the orders are **1, 2, 3, 4, 6, 12**, necessary only. 2

Before the exam ✓

- ✓ Decide nPr vs nCr from the word "order"
- ✓ Write out the six pairwise PIE terms
- ✓ Build the characteristic equation before any algebra
- ✓ "Field = domain + every non-zero invertible"